

Lab 3: Component Modelling & Architectural Pattern Selection

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System	Blood Bank Inventory & Emergency Donor Matcher
Domain	Healthcare & Telemedicine

Architecture Selection

We chose Microservices Architecture for the Blood Bank Inventory & Emergency Donor Matcher system.

Why we chose it

The system has a few different jobs that need to work together, such as keeping track of available blood, finding suitable donors, handling emergency requests, and sending notifications. Keeping these functions as separate services makes the system easier to manage and allows each part to be improved independently.

A second reason is that emergency requests may increase suddenly. The Emergency Donor Matcher can be given more resources when demand is high without having to scale the entire system. Also, if the notification service has a temporary problem, the main inventory and matching functions can continue working.

Security Advantage

Since donor information is sensitive, the Donor Registry can be kept behind a separate service boundary. Other components can access only the information needed for matching, while authenticated APIs and role-based access can control who is allowed to view or update data.

Performance Benefit

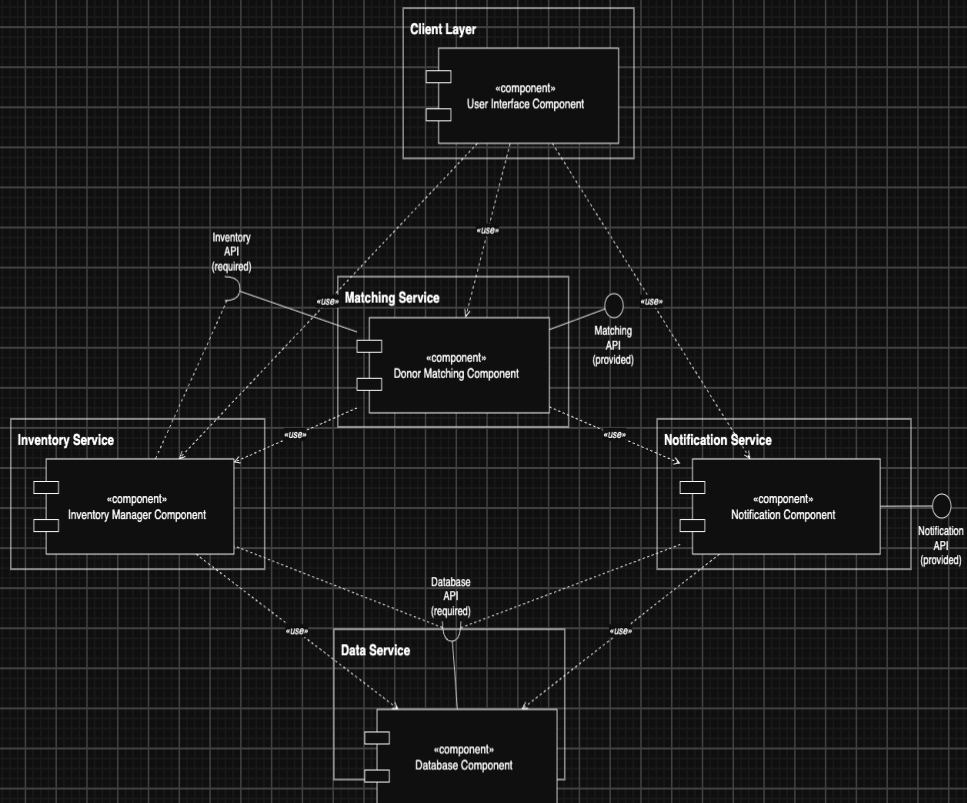
The donor-matching service can be optimized and scaled separately when many emergency requests arrive. Frequently requested blood inventory information can also be cached, which can reduce database access and help the system respond faster.

Main Components

The five main components are Emergency Request Manager, Blood Inventory Manager, Emergency Donor Matcher, Donor Registry, and Notification Service. The system also uses separate Blood Inventory and Donor Registry databases.

UML Diagram

Blood Bank Inventory & Emergency Donor Matcher - Microservices Architecture



Actors: Emergency Requester and Blood Bank Manager interact through the User Interface Component (touch/web/app).

Ball = Provided Interface, Hollow circle = Required Interface, Dashed arrow = «use» dependency.